

Computational Fluid Mechanics: Nonlinear Capillary-Rise Modelling with Finite Differences and Newton-Raphson

Consider an air-liquid interface near a vertical surface, as shown in Figure 1. The liquid wets the surface forming an angle θ , which is called the contact angle. The height of the free surface $y(x)$ is given by the relation:

$$P_1 - P_2 = -2\mathcal{H}\gamma = -\frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}\gamma, \quad (1)$$

where P_1 and P_2 are the pressures of the liquid and air, respectively, $\mathcal{H}$ is the mean curvature of the free surface and γ is the surface tension. The pressure in the gas phase may be considered constant, $P_2 = P_0$, and the hydrostatic pressure in the liquid phase is calculated from $P_1 = P_0 - \rho_L g y$, where ρ_L is the density of the liquid phase and g is the gravitational acceleration. Therefore, the pressures of the gas and liquid phases are equal when $y = 0$ and the interface there becomes horizontal. Substituting the pressure expressions into relation (1), we obtain:

$$\frac{\rho_L g}{\gamma} \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} y = \frac{d^2y}{dx^2}. \quad (2)$$

The characteristic length for this problem, $\lambda = (\gamma/(\rho_L g))^{1/2}$, is called the capillary length and is used to nondimensionalize the height y and the horizontal coordinate x of the free surface. If we define the dimensionless variables $\hat{y} = y/\lambda$ and $\hat{x} = x/\lambda$, then (2) becomes:

$$\left[1 + \left(\frac{d\hat{y}}{d\hat{x}}\right)^2\right]^{\frac{3}{2}} \hat{y} = \frac{d^2\hat{y}}{d\hat{x}^2}. \quad (3)$$

(a) Assume that at a dimensionless distance $\hat{x} = 10$ from the vertical surface the height of the liquid/gas interface is $y = 0$. Find the appropriate expression for the boundary condition that applies at the vertical surface and solve numerically the nonlinear, nondimensional equation (3) using the Newton-Raphson method. Discretize the spatial derivatives by the finite-difference method. Calculate and plot the interface height $y(x)$ when the contact angle is: 1) $\theta = 60$ degrees (hydrophilic surface) and 2) $\theta = 110$ degrees (hydrophobic surface). Perform a sensitivity analysis with respect to the number of discretization nodes.

(b) Consider the case where the liquid is between two vertical surfaces, as shown in Figure 2. Repeat the calculations of question (a) for a hydrophobic and a hydrophilic surface, assuming that the dimensionless distance between the two vertical surfaces is $L = 2, 4, 6$ and 8 . How does the catenary rise of the liquid, c , the interface height at the midpoint between the two vertical walls, vary as a function of the distance L between the vertical walls? Plot this relation and examine whether there is a correlation between the catenary rise and the distance L .

(c) Solve equation (3) analytically, assuming that the magnitude of the interface slope is very small, i.e. $(d\hat{y}/d\hat{x})^2 \ll 1$. Apply the boundary conditions for the case examined in question (b) and compare the analytical solution with the numerical solution of the problem when the distance between the two surfaces is $L = 2$ and the contact angle is $\theta = 85, 95$ and 130 degrees.

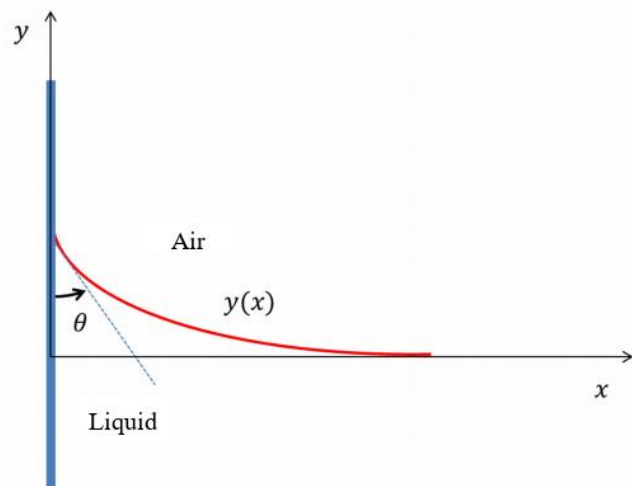


Figure 1. Liquid-air interface near a vertical surface.

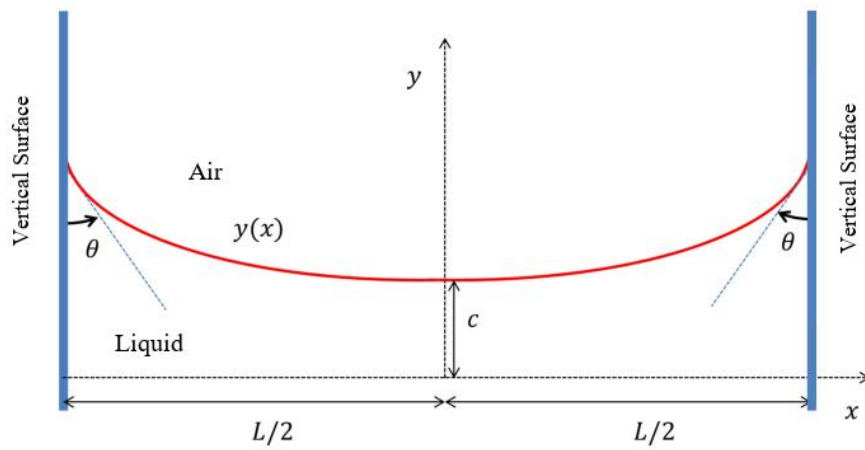


Figure 2. Liquid-air interface between two vertical surfaces and catenary rise c .

The problem to be solved is the boundary-value problem expressed by the following relations:

$$\frac{\rho_L g}{L} \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{3}{2}} y = \frac{d^2 y}{dx^2} \quad (1)$$

$$\left. \frac{dy}{dx} \right|_{x=0} = -\cot(\theta) \quad (2)$$

$$y(L) = 0 \quad (3)$$

which are expressed in dimensionless form, as defined, as:

$$\left[1 + \left(\frac{d\hat{y}}{d\hat{x}} \right)^2 \right]^{\frac{3}{2}} \hat{y} = \frac{d^2 \hat{y}}{d\hat{x}^2} \quad (4)$$

$$\left. \frac{d\hat{y}}{d\hat{x}} \right|_{\hat{x}=0} = -\cot(\theta) \quad (5)$$

$$\hat{y}(10) = 0 \quad (6)$$

where $\cot(\theta)$ is the cotangent of the liquid contact angle with the vertical surface. The solution of equation (4), subject to boundary conditions (5) and (6), is carried out by the finite-difference method. Specifically, the interval $\hat{x}$ in $[0,10]$ is discretized into n equally spaced nodes with spacing h , where the value of $\hat{y}$ is calculated. The first and second derivatives at each node are approximated by finite differences as follows:

$$\frac{d\hat{y}}{d\hat{x}} \approx \frac{\hat{y}_{i+1} - \hat{y}_{i-1}}{2h} \quad (7)$$

$$\frac{d^2 \hat{y}}{d\hat{x}^2} \approx \frac{\hat{y}_{i+1} - 2\hat{y}_i + \hat{y}_{i-1}}{h^2} \quad (8)$$

Substituting relations (7) and (8) into equation (4) and expanding the latter, one obtains for each node an equation of the form:

$$8h(\hat{y}_{i+1} - 2\hat{y}_i + \hat{y}_{i-1}) - \hat{y}_i(4h^2 + \hat{y}_{i+1}^2 - 2\hat{y}_{i+1}\hat{y}_{i-1} + \hat{y}_{i-1}^2)^{\frac{3}{2}} = 0 \quad (9)$$

while following the same procedure for relations (5) and (6), the following relations are obtained:

$$\hat{y}_0 = \hat{y}_2 + 2h \cot(\theta) \quad (10)$$

$$\hat{y}_n = 0 \quad (11)$$

Defining the vector $\hat{y}$ of dimension n , which contains all elements $\hat{y}_i$ for i in $\{1, \dots, n\}$, relation (9) yields n equations of the form $f_i(\hat{y})$, which constitute the n -dimensional vector $f(\hat{y})$. Therefore, a system of n nonlinear equations with n unknowns is defined, which can be solved numerically by the Newton-Raphson method, according to which each new estimate of vector $\hat{y}$ is calculated as:

$$\delta \hat{\mathbf{y}}^{(k)} = -(\mathbf{J}^{(k)})^{-1} \mathbf{f}(\hat{\mathbf{y}}^{(k)}) \quad (12)$$

$$\hat{\mathbf{y}}^{(k+1)} = \hat{\mathbf{y}}^{(k)} + \delta \hat{\mathbf{y}}^{(k)} \quad (13)$$

where $\mathbf{J}$ is the Jacobian matrix, each element of which is defined as:

$$J_{ij} = \frac{\partial f_i(\hat{\mathbf{y}})}{\partial \hat{y}_j} \quad (14)$$

Based on relations (9)-(11), the expressions for each element of vector $\mathbf{f}(\hat{\mathbf{y}})$ are obtained as:

$$f_1(\hat{\mathbf{y}}) = 2(\hat{y}_2 - \hat{y}_1 + h \cot(\theta)) - h^2 \hat{y}_1 (1 + \cot^2(\theta))^{\frac{3}{2}} \quad (15)$$

$$f_i(\hat{\mathbf{y}}) = 8h(\hat{y}_{i+1} - 2\hat{y}_i + \hat{y}_{i-1}) - \hat{y}_i(4h^2 + \hat{y}_{i+1}^2 - 2\hat{y}_{i+1}\hat{y}_{i-1} + \hat{y}_{i-1}^2)^{\frac{3}{2}}, i \in \{2, \dots, n-1\} \quad (16)$$

$$f_n(\hat{\mathbf{y}}) = \hat{y}_n - 0 \quad (17)$$

Therefore, the elements of the Jacobian matrix can be calculated analytically as:

$$J_{1,1} = -2 - h^2(1 + \cot^2(\theta))^{\frac{3}{2}} \quad (18)$$

$$J_{1,2} = 2 \quad (19)$$

$$J_{i,i-1} = 8h - 3\hat{y}_i(4h^2 + \hat{y}_{i+1}^2 - 2\hat{y}_{i+1}\hat{y}_{i-1} + \hat{y}_{i-1}^2)^{\frac{1}{2}}(\hat{y}_{i-1} - \hat{y}_{i+1}), i \in \{2, \dots, n-1\} \quad (20)$$

$$J_{i,i} = -16h - (4h^2 + \hat{y}_{i+1}^2 - 2\hat{y}_{i+1}\hat{y}_{i-1} + \hat{y}_{i-1}^2)^{\frac{3}{2}}, i \in \{2, \dots, n-1\} \quad (21)$$

$$J_{i,i+1} = 8h - 3\hat{y}_i(4h^2 + \hat{y}_{i+1}^2 - 2\hat{y}_{i+1}\hat{y}_{i-1} + \hat{y}_{i-1}^2)^{\frac{1}{2}}(\hat{y}_{i+1} - \hat{y}_{i-1}), i \in \{2, \dots, n-1\} \quad (22)$$

$$J_{nn} = 1 \quad (23)$$

where the elements not appearing in equations (18)-(23) are zero. The results of solving the boundary-value problem for 31 nodes and for angles $\theta = 60$ degrees and $\theta = 110$ degrees are given in the table below:

Table 1: Results of the boundary-value problem

Node No.	Dimensionless distance	Dimensionless interface height	
		$\theta = 60$ deg	$\theta = 110$ deg
1	0	5.01e-01	-3.40e-01
2	0.333	3.51e-01	-2.42e-01
3	0.667	2.49e-01	-1.73e-01
4	1	1.78e-01	-1.23e-01
5	1.333	1.27e-01	-8.85e-02
6	1.667	9.13e-02	-6.35e-02
7	2	6.55e-02	-4.55e-02
8	2.333	4.70e-02	-3.27e-02
9	2.667	3.37e-02	-2.34e-02

10	3	2.42e-02	-1.68e-02
11	3.333	1.73e-02	-1.21e-02
12	3.667	1.24e-02	-8.66e-03
13	4	8.93e-03	-6.22e-03
14	4.333	6.41e-03	-4.46e-03
15	4.667	4.60e-03	-3.20e-03
16	5	3.30e-03	-2.30e-03
17	5.333	2.37e-03	-1.65e-03
18	5.667	1.70e-03	-1.18e-03
19	6	1.22e-03	-8.49e-04
20	6.333	8.75e-04	-6.09e-04
21	6.667	6.27e-04	-4.37e-04
22	7	4.50e-04	-3.13e-04
23	7.333	3.22e-04	-2.24e-04
24	7.667	2.30e-04	-1.60e-04
25	8	1.64e-04	-1.14e-04
26	8.333	1.15e-04	-8.02e-05
27	8.667	7.98e-05	-5.55e-05
28	9	5.32e-05	-3.70e-05
29	9.333	3.25e-05	-2.26e-05
30	9.667	1.54e-05	-1.07e-05
31	10	0	0

The results of Table 1 are plotted below, together with the solutions computed for different numbers of nodes.

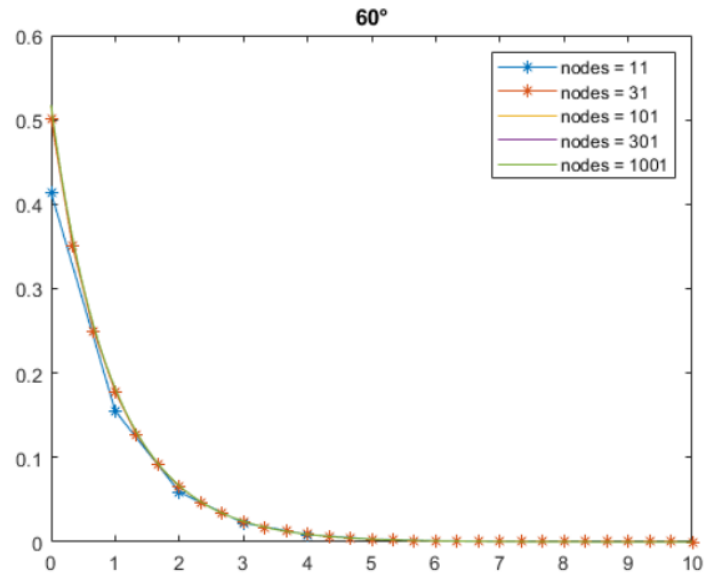


Figure 1: Solution of the boundary-value problem for $\theta = 60$ degrees

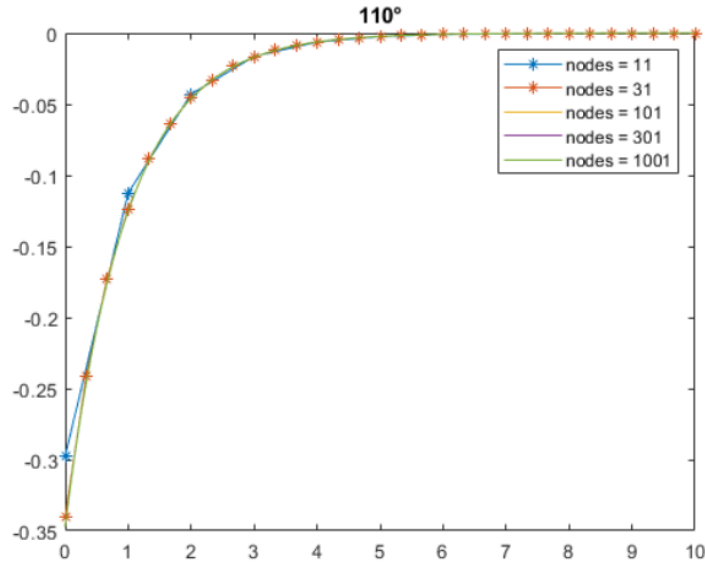


Figure 2: Solution of the boundary-value problem for $\theta = 110$ degrees

In order to perform a sensitivity analysis of the solution with respect to the number of nodes, the boundary-value problem is solved each time for a different number of nodes and the variation of the following norm is examined:

$$s_i = \sqrt{\sum_{k=1}^{n_c} (\hat{y}_{i,k} - \hat{y}_{i-1,k})^2} \quad (24)$$

where index i refers to the solutions with the i -th set of nodes, while index k refers to the common nodes between the different node sets, where n_c is the number of common nodes. The values of norm s_i for different numbers of nodes are given in the table below:

Table 2: Values of norm s_i for different numbers of nodes

Number of nodes	Norm s_i	
	$\theta = 60$ deg	$\theta = 110$ deg
11	-	-
31	8.80e-02	3.02e-02
101	1.69e-02	4.87e-03
301	1.61e-03	4.50e-04
1001	1.84e-04	5.14e-05

Based on the results of Table 2, increasing the number of nodes leads to a decrease in norm s_i , as the computed solution converges to the true solution due to a more accurate approximation of the derivatives from relations (7) and (8).

For checking the quadratic convergence of the Newton-Raphson method, the variation of the norm is examined:

$$\|\delta \hat{\mathbf{y}}^{(k)}\| = \sqrt{\sum_i^n |\delta \hat{y}_i^{(k)}|^2} \quad (25)$$

as a function of the number of iterations k . This check is performed for the case of the solution with 31 nodes, with the results listed in the table below:

Table 3: Values of norm $\|\delta \hat{\mathbf{y}}^{(k)}\|$ as a function of the number of iterations

Number of iterations	Norm $\ \delta \hat{\mathbf{y}}^{(k)}\ $	
	theta = 60 deg	theta = 110 deg
1	7.51e-01	4.99e-01
2	4.43e-02	1.41e-02
3	2.21e-04	1.57e-05
4	1.67e-09	6.43e-12
5	1.45e-16	5.17e-17

From the results of Table 3, the quadratic reduction of the error is evident, as approximated by the norm of relation (25), except for the last iteration.

This code was developed in the MATLAB R2018b programming environment.

```
clear;clc

n = [11, 31, 101, 301, 1001];
y = zeros(length(n)-1,2);

maxiter = 30;
theta = [60, 110];
es = 1.e-12;

x = cell(1, length(n));
u = cell(length(n), length(theta));
f = cell(length(n), length(theta));
J = cell(length(n), length(theta));
du = cell(length(n), length(theta));
ea = cell(length(n), length(theta));

for j = 1:length(theta)
    figure(j)
    for i = 1:length(n)
        uo = zeros(n(i), 1);
        u{i,j} = uo;

        iter = 0;
        while true
            iter = iter + 1;
            [f{i,j}, J{i,j}] = Capillary_ND(u{i,j}, theta(j));
            du{i,j} = -J{i,j}\f{i,j};
            u{i,j} = u{i,j} + du{i,j};
            ea{i,j}(iter) = norm(du{i,j}, 2);
            fprintf('%d %e \n', iter, ea{i,j}(iter))

            if ea{i,j}(iter) <= es || iter >= maxiter; break; end
        end

        ea{i,j} = ea{i,j}';

        if j == 1
            x{i} = linspace(0, 10, n(i));
        end

        if (i >= 1) && (i <= 2)
            plot(x{i}, u{i,j}, '-*')
        else
            plot(x{i}, u{i,j}, '-')
        end
        hold on
    end
    hold off

    if j == 1; title('y = f(x), 60°'); else; title('y = f(x), 110°'); end

    xlabel('x'); ylabel('y')
    legend('nodes = 11', 'nodes = 31', 'nodes = 101', 'nodes = 301', ...
        'nodes = 1001')

    for k = 1:length(n)-1
        for m = 0:10
            y(k,j) = y(k,j) + norm(u{k,j} (1+m*(n(k)-1)/10)^2 - ...
                u{k+1,j} (1+m*(n(k+1)-1)/10)^2, 2);
        end
    end
end
end
```


The boundary-value problem, with respect to boundary condition (6), is expressed as:

$$\left. \frac{dy}{dx} \right|_{x=L} = \cot(\theta) \quad (26)$$

therefore relations (17) and (23) are replaced by:

$$f_n(\mathcal{Y}) = 2(\hat{y}_{n-1} - \hat{y}_n + h \cot(\theta)) - h^2 \hat{y}_n (1 + \cot^2(\theta))^{\frac{3}{2}} \quad (27)$$

$$J_{n,n-1} = 2 \quad (28)$$

$$J_{n,n} = -2 - h^2 (1 + \cot^2(\theta))^{\frac{3}{2}} \quad (29)$$

The results of solving the new boundary-value problem for 31 nodes and angles theta = 60 degrees and theta = 110 degrees are given in the following tables:

Table 4: Results of the boundary-value problem for theta = 60 degrees

Node No.	Dimensionless surface distance			
	L = 2		L = 4	
	Dimensionless distance	Dimensionless interface height	Dimensionless distance	Dimensionless interface height

1	-1	6.68e-01	-2	5.32e-01
2	-0.933	6.31e-01	-1.867	4.63e-01
3	-0.867	5.99e-01	-1.733	4.04e-01
4	-0.8	5.70e-01	-1.6	3.55e-01
5	-0.733	5.45e-01	-1.467	3.13e-01
6	-0.667	5.22e-01	-1.333	2.77e-01
7	-0.6	5.02e-01	-1.2	2.47e-01
8	-0.533	4.85e-01	-1.067	2.21e-01
9	-0.467	4.70e-01	-0.933	1.99e-01
10	-0.4	4.57e-01	-0.8	1.82e-01
11	-0.333	4.46e-01	-0.667	1.67e-01
12	-0.267	4.38e-01	-0.533	1.55e-01
13	-0.2	4.31e-01	-0.4	1.47e-01
14	-0.133	4.26e-01	-0.267	1.41e-01
15	-0.067	4.24e-01	-0.133	1.37e-01
16	0	4.23e-01	0	1.36e-01
17	0.067	4.24e-01	0.133	1.37e-01
18	0.133	4.26e-01	0.267	1.41e-01
19	0.2	4.31e-01	0.4	1.47e-01
20	0.267	4.38e-01	0.533	1.55e-01
21	0.333	4.46e-01	0.667	1.67e-01
22	0.4	4.57e-01	0.8	1.82e-01
23	0.467	4.70e-01	0.933	1.99e-01
24	0.533	4.85e-01	1.067	2.21e-01
25	0.6	5.02e-01	1.2	2.47e-01
26	0.667	5.22e-01	1.333	2.77e-01
27	0.733	5.45e-01	1.467	3.13e-01
28	0.8	5.70e-01	1.6	3.55e-01
29	0.867	5.99e-01	1.733	4.04e-01
30	0.933	6.31e-01	1.867	4.63e-01
31	1	6.68e-01	2	5.32e-01

Continuation of Table 4

Node No.	Dimensionless surface distance			
	L = 6		L = 8	
	Dimensionless distance	Dimensionless interface height	Dimensionless distance	Dimensionless interface height
1	-3	5.14e-01	-4	5.07e-01
2	-2.8	4.14e-01	-3.733	3.81e-01
3	-2.6	3.36e-01	-3.467	2.89e-01
4	-2.4	2.74e-01	-3.2	2.20e-01
5	-2.2	2.24e-01	-2.933	1.68e-01
6	-2	1.84e-01	-2.667	1.29e-01
7	-1.8	1.52e-01	-2.4	9.91e-02
8	-1.6	1.26e-01	-2.133	7.63e-02
9	-1.4	1.05e-01	-1.867	5.91e-02

10	-1.2	8.85e-02	-1.6	4.60e-02
11	-1	7.54e-02	-1.333	3.62e-02
12	-0.8	6.54e-02	-1.067	2.91e-02
13	-0.6	5.79e-02	-0.8	2.39e-02
14	-0.4	5.29e-02	-0.533	2.05e-02
15	-0.2	4.99e-02	-0.267	1.86e-02
16	0	4.89e-02	0	1.79e-02
17	0.2	4.99e-02	0.267	1.86e-02
18	0.4	5.29e-02	0.533	2.05e-02
19	0.6	5.79e-02	0.8	2.39e-02
20	0.8	6.54e-02	1.067	2.91e-02
21	1	7.54e-02	1.333	3.62e-02
22	1.2	8.85e-02	1.6	4.60e-02
23	1.4	1.05e-01	1.867	5.91e-02
24	1.6	1.26e-01	2.133	7.63e-02
25	1.8	1.52e-01	2.4	9.91e-02
26	2	1.84e-01	2.667	1.29e-01
27	2.2	2.24e-01	2.933	1.68e-01
28	2.4	2.74e-01	3.2	2.20e-01
29	2.6	3.36e-01	3.467	2.89e-01
30	2.8	4.14e-01	3.733	3.81e-01
31	3	5.14e-01	4	5.07e-01

Table 5: Results of the boundary-value problem for $\theta = 110$ degrees

Node No.	Dimensionless surface distance			
	L = 2		L = 4	
	Dimensionless distance	Dimensionless interface height	Dimensionless distance	Dimensionless interface height
1	-1	-4.52e-01	-2	-3.59e-01
2	-0.933	-4.29e-01	-1.867	-3.14e-01
3	-0.867	-4.08e-01	-1.733	-2.76e-01
4	-0.8	-3.90e-01	-1.6	-2.43e-01
5	-0.733	-3.73e-01	-1.467	-2.15e-01
6	-0.667	-3.58e-01	-1.333	-1.90e-01
7	-0.6	-3.44e-01	-1.2	-1.70e-01
8	-0.533	-3.33e-01	-1.067	-1.52e-01
9	-0.467	-3.22e-01	-0.933	-1.37e-01
10	-0.4	-3.14e-01	-0.8	-1.25e-01
11	-0.333	-3.06e-01	-0.667	-1.15e-01
12	-0.267	-3.01e-01	-0.533	-1.07e-01
13	-0.2	-2.96e-01	-0.4	-1.01e-01
14	-0.133	-2.93e-01	-0.267	-9.70e-02
15	-0.067	-2.91e-01	-0.133	-9.45e-02
16	0	-2.90e-01	0	-9.36e-02
17	0.067	-2.91e-01	0.133	-9.45e-02
18	0.133	-2.93e-01	0.267	-9.70e-02

19	0.2	-2.96e-01	0.4	-1.01e-01
20	0.267	-3.01e-01	0.533	-1.07e-01
21	0.333	-3.06e-01	0.667	-1.15e-01
22	0.4	-3.14e-01	0.8	-1.25e-01
23	0.467	-3.22e-01	0.933	-1.37e-01
24	0.533	-3.33e-01	1.067	-1.52e-01
25	0.6	-3.44e-01	1.2	-1.70e-01
26	0.667	-3.58e-01	1.333	-1.90e-01
27	0.733	-3.73e-01	1.467	-2.15e-01
28	0.8	-3.90e-01	1.6	-2.43e-01
29	0.867	-4.08e-01	1.733	-2.76e-01
30	0.933	-4.29e-01	1.867	-3.14e-01
31	1	-4.52e-01	2	-3.59e-01

Continuation of Table 5

Node No.	Dimensionless surface distance			
	L = 6		L = 8	
	Dimensionless distance	Dimensionless interface height	Dimensionless distance	Dimensionless interface height
1	-3	-3.46e-01	-4	-3.43e-01
2	-2.8	-2.82e-01	-3.733	-2.61e-01
3	-2.6	-2.30e-01	-3.467	-1.99e-01
4	-2.4	-1.88e-01	-3.2	-1.52e-01
5	-2.2	-1.55e-01	-2.933	-1.16e-01
6	-2	-1.27e-01	-2.667	-8.94e-02
7	-1.8	-1.05e-01	-2.4	-6.87e-02
8	-1.6	-8.71e-02	-2.133	-5.29e-02
9	-1.4	-7.27e-02	-1.867	-4.10e-02
10	-1.2	-6.12e-02	-1.6	-3.19e-02
11	-1	-5.22e-02	-1.333	-2.51e-02
12	-0.8	-4.52e-02	-1.067	-2.02e-02
13	-0.6	-4.01e-02	-0.8	-1.66e-02
14	-0.4	-3.66e-02	-0.533	-1.42e-02
15	-0.2	-3.45e-02	-0.267	-1.29e-02
16	0	-3.38e-02	0	-1.24e-02
17	0.2	-3.45e-02	0.267	-1.29e-02
18	0.4	-3.66e-02	0.533	-1.42e-02
19	0.6	-4.01e-02	0.8	-1.66e-02
20	0.8	-4.52e-02	1.067	-2.02e-02
21	1	-5.22e-02	1.333	-2.51e-02
22	1.2	-6.12e-02	1.6	-3.19e-02
23	1.4	-7.27e-02	1.867	-4.10e-02
24	1.6	-8.71e-02	2.133	-5.29e-02
25	1.8	-1.05e-01	2.4	-6.87e-02
26	2	-1.27e-01	2.667	-8.94e-02
27	2.2	-1.55e-01	2.933	-1.16e-01

28	2.4	-1.88e-01	3.2	-1.52e-01
29	2.6	-2.30e-01	3.467	-1.99e-01
30	2.8	-2.82e-01	3.733	-2.61e-01
31	3	-3.46e-01	4	-3.43e-01

The results of Table 4 and Table 5 are plotted below.

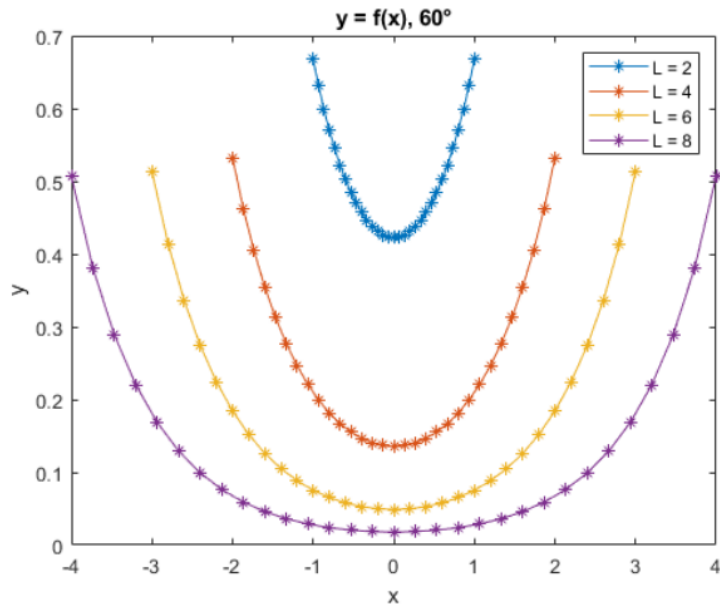


Figure 3: Solution of the boundary-value problem for $\theta = 60$ degrees

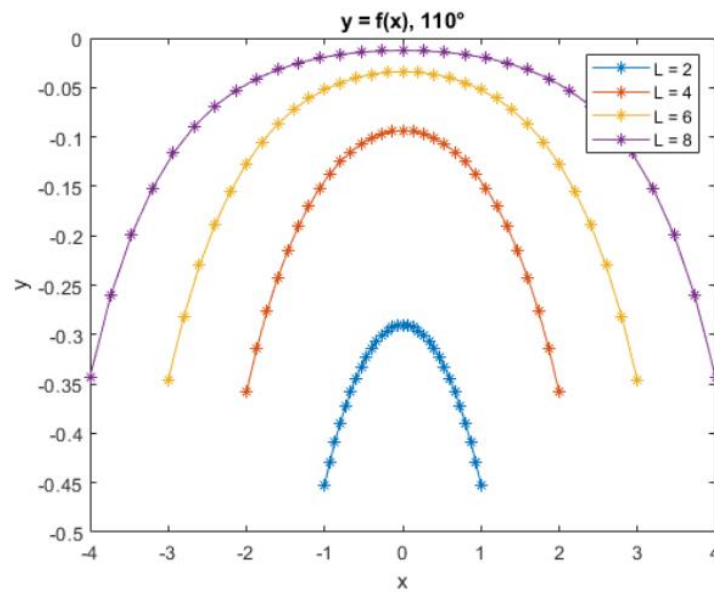


Figure 4: Solution of the boundary-value problem for $\theta = 110$ degrees

The catenary rise (or fall) of the liquid for the two contact angles is shown in the following diagrams:

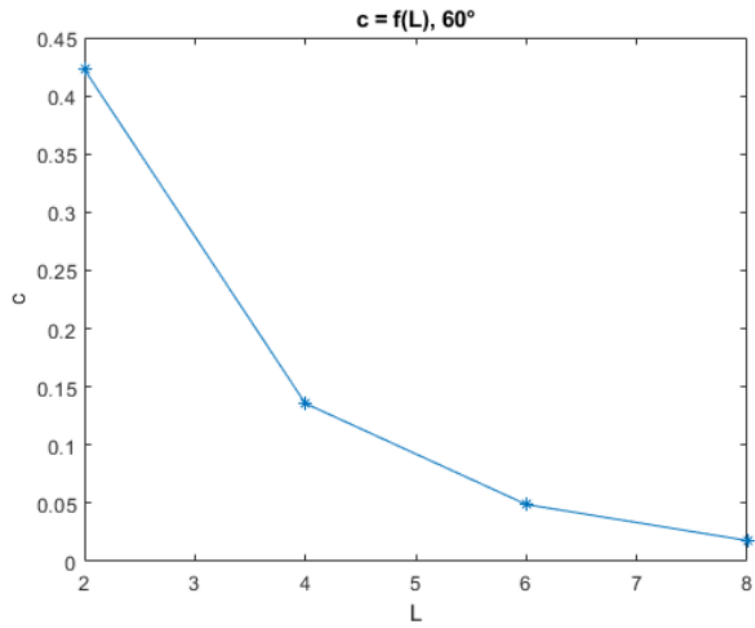


Figure 5: Plot of the catenary rise c as a function of distance L between the two vertical surfaces for $\theta = 60$ degrees

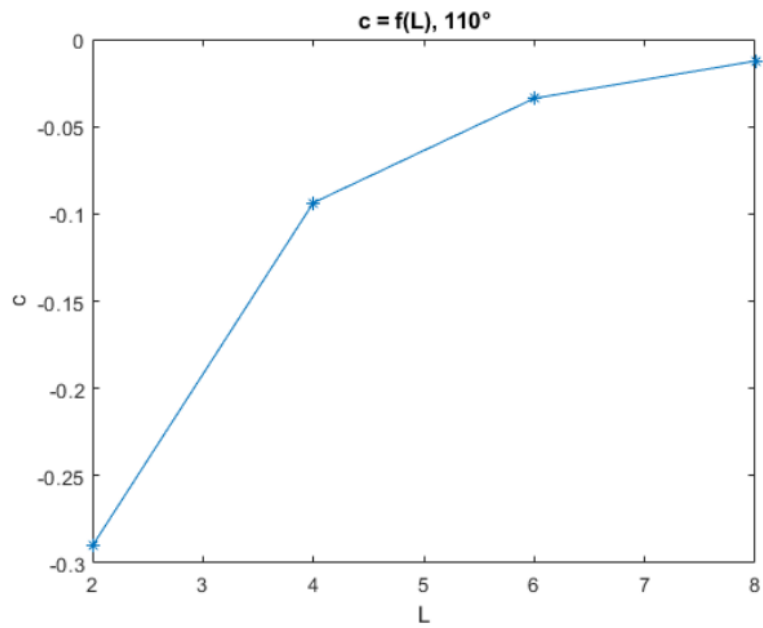


Figure 6: Plot of the catenary descent c as a function of distance L between the two vertical surfaces for $\theta = 110$ degrees

```

clear;clc

n = 31;

maxiter = 30;
L = [2; 4; 6; 8];
theta = [60, 110];
es = 1.e-12;
uo = zeros(n, 1);

x = cell(length(L), 1);
u = cell(length(L), length(theta));
f = cell(length(L), length(theta));
J = cell(length(L), length(theta));
du = cell(length(L), length(theta));

c = zeros(length(L), length(theta));

for j = 1:length(theta)
    figure(j)
    for i = 1:length(L)
        u{i,j} = uo;

        iter = 0;
        while true
            iter = iter + 1;
            [f{i,j}, J{i,j}] = Capillary_NN(u{i,j}, L(i), theta(j));
            du{i,j} = -J{i,j}\f{i,j};
            u{i,j} = u{i,j} + du{i,j};
            ea = norm(du{i,j}, 2);
            fprintf('%d %e \n', iter, ea)
            if ea <= es || iter >= maxiter; break; end
        end

        if j == 1
            x{i} = linspace(-L(i)/2, L(i)/2, n);
        end

        plot(x{i}, u{i,j}, '-*')
        hold on
    end
    hold off
    if j == 1; title('y = f(x), 60°'); else; title('y = f(x), 110°'); end
    xlabel('x'); ylabel('y')
    legend('L = 2', 'L = 4', 'L = 6', 'L = 8')
end

for j = 1:length(theta)
    for i = 1:length(L)
        c(i,j) = u{i,j}((n+1)/2);
    end
    figure(length(theta)+j)
    plot(L, c(:,j), '-*')
    if j == 1; title('c = f(L), 60°'); else; title('c = f(L), 110°'); end
    xlabel('L'); ylabel('c')
end

```


Assuming $(dy/dx)^2 \ll 1$, equation (4) is written as:

$$\hat{y} = \frac{d^2 \hat{y}}{d\hat{x}^2} \quad (30)$$

whose solution, given boundary conditions (5) and (26), is:

$$\hat{y} = \frac{e^L + 1}{e^{2L} - 1} (e^{L-x} + e^x) \quad (31)$$

The interface-height results calculated analytically for $L = 2$ from relation (31) are compared with those calculated numerically from the boundary-value problem of equations (4), (5) and (26) in the tables below:

Table 6: Interface height: comparison of the approximate analytical solution with the numerical solution

Node No.	Dimensionless distance	Dimensionless interface height			
		theta = 85 deg		theta = 95 deg	
		Analytical	Numerical	Analytical	Numerical
1	0.000	1.15e-01	1.14e-01	-1.15e-01	-1.14e-01
2	0.067	1.09e-01	1.09e-01	-1.09e-01	-1.09e-01
3	0.133	1.04e-01	1.04e-01	-1.04e-01	-1.04e-01
4	0.200	9.96e-02	9.92e-02	-9.96e-02	-9.92e-02
5	0.267	9.54e-02	9.50e-02	-9.54e-02	-9.50e-02
6	0.333	9.16e-02	9.12e-02	-9.16e-02	-9.12e-02
7	0.400	8.83e-02	8.79e-02	-8.83e-02	-8.79e-02
8	0.467	8.53e-02	8.49e-02	-8.53e-02	-8.49e-02
9	0.533	8.27e-02	8.23e-02	-8.27e-02	-8.23e-02
10	0.600	8.05e-02	8.01e-02	-8.05e-02	-8.01e-02
11	0.667	7.86e-02	7.83e-02	-7.86e-02	-7.83e-02
12	0.733	7.71e-02	7.68e-02	-7.71e-02	-7.68e-02
13	0.800	7.59e-02	7.56e-02	-7.59e-02	-7.56e-02
14	0.867	7.51e-02	7.48e-02	-7.51e-02	-7.48e-02
15	0.933	7.46e-02	7.43e-02	-7.46e-02	-7.43e-02
16	1.000	7.44e-02	7.41e-02	-7.44e-02	-7.41e-02
17	1.067	7.46e-02	7.43e-02	-7.46e-02	-7.43e-02
18	1.133	7.51e-02	7.48e-02	-7.51e-02	-7.48e-02
19	1.200	7.59e-02	7.56e-02	-7.59e-02	-7.56e-02
20	1.267	7.71e-02	7.68e-02	-7.71e-02	-7.68e-02

21	1.333	7.86e-02	7.83e-02	-7.86e-02	-7.83e-02
22	1.400	8.05e-02	8.01e-02	-8.05e-02	-8.01e-02
23	1.467	8.27e-02	8.23e-02	-8.27e-02	-8.23e-02
24	1.533	8.53e-02	8.49e-02	-8.53e-02	-8.49e-02
25	1.600	8.83e-02	8.79e-02	-8.83e-02	-8.79e-02
26	1.667	9.16e-02	9.12e-02	-9.16e-02	-9.12e-02
27	1.733	9.54e-02	9.50e-02	-9.54e-02	-9.50e-02
28	1.800	9.96e-02	9.92e-02	-9.96e-02	-9.92e-02
29	1.867	1.04e-01	1.04e-01	-1.04e-01	-1.04e-01
30	1.933	1.09e-01	1.09e-01	-1.09e-01	-1.09e-01
31	2.000	1.15e-01	1.14e-01	-1.15e-01	-1.14e-01

Continuation of Table 6

Node No.	Dimensionless distance	Dimensionless interface height	
		theta = 130 deg	
		Analytical	Numerical
1	0.000	-8.70e-01	-1.10e+00
2	0.067	-8.18e-01	-1.05e+00
3	0.133	-7.73e-01	-9.99e-01
4	0.200	-7.34e-01	-9.55e-01
5	0.267	-7.00e-01	-9.15e-01
6	0.333	-6.70e-01	-8.79e-01
7	0.400	-6.44e-01	-8.46e-01
8	0.467	-6.21e-01	-8.18e-01
9	0.533	-6.01e-01	-7.93e-01
10	0.600	-5.85e-01	-7.72e-01
11	0.667	-5.71e-01	-7.54e-01
12	0.733	-5.60e-01	-7.40e-01
13	0.800	-5.51e-01	-7.28e-01
14	0.867	-5.45e-01	-7.20e-01
15	0.933	-5.42e-01	-7.16e-01
16	1.000	-5.40e-01	-7.14e-01
17	1.067	-5.42e-01	-7.16e-01
18	1.133	-5.45e-01	-7.20e-01
19	1.200	-5.51e-01	-7.28e-01
20	1.267	-5.60e-01	-7.40e-01
21	1.333	-5.71e-01	-7.54e-01
22	1.400	-5.85e-01	-7.72e-01
23	1.467	-6.01e-01	-7.93e-01
24	1.533	-6.21e-01	-8.18e-01
25	1.600	-6.44e-01	-8.46e-01
26	1.667	-6.70e-01	-8.79e-01
27	1.733	-7.00e-01	-9.15e-01
28	1.800	-7.34e-01	-9.55e-01
29	1.867	-7.73e-01	-9.99e-01
30	1.933	-8.18e-01	-1.05e+00
31	2.000	-8.70e-01	-1.10e+00

The results of Table 6 are plotted in the following diagrams:

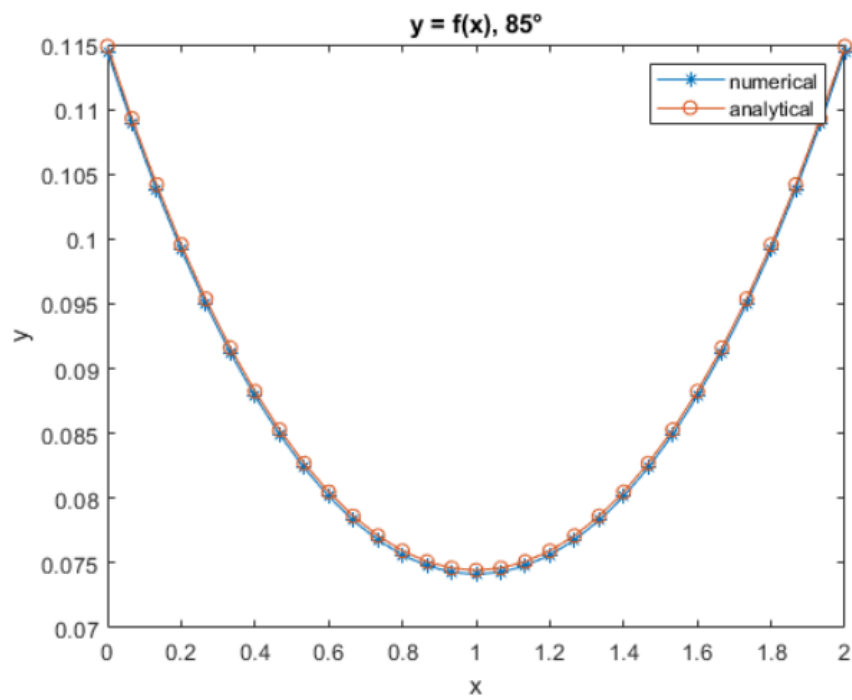


Figure 7: Approximate analytical solution and numerical solution of the boundary-value problem for $\theta = 85$ degrees

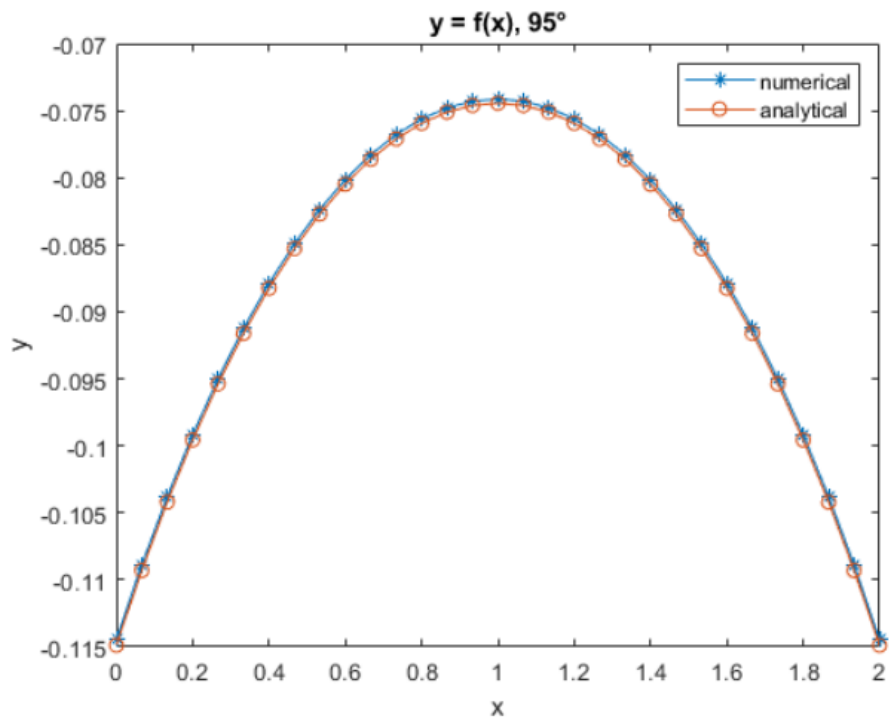


Figure 8: Approximate analytical solution and numerical solution of the boundary-value problem for $\theta = 95$ degrees

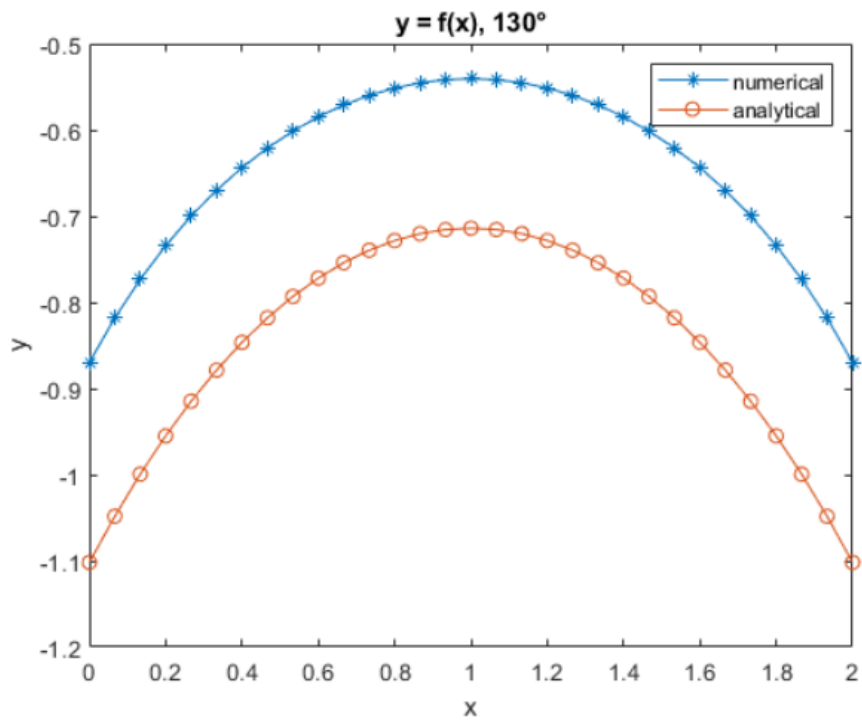


Figure 9: Approximate analytical solution and numerical solution of the boundary-value problem for $\theta = 130$ degrees

From the results of Table 6 and Figures 7-9, it is evident that the assumption $(dy/dx)^2 \ll 1$ is accurate only for contact angles close to 90 degrees, where the catenary-rise phenomenon is weak and the derivative dy/dx indeed tends to small values over the whole domain.

```
clear;clc

n = 31;

maxiter = 30;
L = 2;
theta = [85, 95, 130];
es = 1.e-12;
uo = zeros(n, 1);

u = zeros(n, length(theta));
ua = zeros(n, length(theta));
f = zeros(n, length(theta));
J = cell(length(theta),1);
du = zeros(n, length(theta));

x = linspace(0, L, n);
for j = 1:length(theta)
    u(:,j) = uo;

    iter = 0;
    while true
        iter = iter + 1;
        [f(:,j), J{j}] = Capillary_NN(u(:,j), L, theta(j));
        du(:,j) = -J{j}\f(:,j);
        u(:,j) = u(:,j) + du(:,j);
        ea = norm(du(:,j), 2);
        fprintf('%d %e \n', iter, ea)
        if ea <= es || iter >= maxiter; break; end
    end

    ua(:,j) = (exp(L)+1)*(exp(L-x)+exp(x)) ...
        / (exp(2*L)-1)*cot(theta(j)/180*pi);

    figure(j)
    plot(x, u(:,j), '-*')
    hold on
    plot(x, ua(:,j), '-o')
    hold off
    if j == 1
        title('y = f(x), 85°')
    elseif j == 2
        title('y = f(x), 95°')
    else
        title('y = f(x), 130°')
    end
    xlabel('x'); ylabel('y')
    legend('numerical', 'analytical')
end
```

Capillary_ND.m

```
function [f, J] = Capillary_ND(u, theta)

n = length(u);
a = 0; b = 10;
h = (b-a)/(n-1);

f = zeros(n,1);

for i = 1:n
    if i == 1
        f(i) = 2*(u(i+1)-u(i)+h*cot(theta/180*pi)) ...
            - h^2*u(i)*(1+cot(theta/180*pi)^2)^(3/2);
    elseif i == n
        f(i) = u(i)-0;
    else
        f(i) = 8*h*(u(i+1)-2*u(i)+u(i-1)) ...
            - u(i)*(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(3/2);
    end
end

J = zeros(n,n);

J(1,1) = -2-h^2*(1+cot(theta/180*pi)^2)^(3/2);
J(1,2) = 2;

for i = 2:n-1

    J(i,i-1) = 8*h- ...
        3*u(i)*(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(1/2) ...
        *(u(i-1)-u(i+1));
    J(i,i) = -16*h-(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(3/2);
    J(i,i+1) = 8*h- ...
        3*u(i)*(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(1/2) ...
        *(u(i+1)-u(i-1));

end

J(n,n) = 1;
```

Capillary_NN.m

```
function [f, J] = Capillary_NN(u, b, theta)

n = length(u);
a = 0;
h = (b-a)/(n-1);

f = zeros(n,1);

for i = 1:n
    if i == 1
        f(i) = 2*(u(i+1)-u(i)+h*cot(theta/180*pi)) ...
            - h^2*u(i)*(1+cot(theta/180*pi)^2)^(3/2);
    elseif i == n
        f(i) = 2*(u(i-1)-u(i)+h*cot(theta/180*pi)) ...
            - h^2*u(i)*(1+cot(theta/180*pi)^2)^(3/2);
    else
        f(i) = 8*h*(u(i+1)-2*u(i)+u(i-1)) ...
            - u(i)*(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(3/2);
    end
end

J = zeros(n,n);

J(1,1) = -2*h^2*(1+cot(theta/180*pi)^2)^(3/2);
J(1,2) = 2;

for i = 2:n-1
    J(i,i-1) = 8*h- ...
        3*u(i)*(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(1/2) ...
        *(u(i-1)-u(i+1));
    J(i,i) = -16*h-(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(3/2);
    J(i,i+1) = 8*h- ...
        3*u(i)*(4*h^2+u(i+1)^2-2*u(i+1)*u(i-1)+u(i-1)^2)^(1/2) ...
        *(u(i+1)-u(i-1));
end

J(n,n-1) = 2;
J(n,n) = -2*h^2*(1+cot(theta/180*pi)^2)^(3/2);
```